

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

T. M. Nestor

- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

T. M. Nestor

Independent researcher, Australia

Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells coupled through the background Green's tensor: the elastic coupled-dipole method. Because cubes tile space, n planes of voxels of side D/n are the layer itself for every n , with no shape error, so the difference from the exact solution is the discretisation error alone. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying Legendre moments of its internal field to degree p converges at order $2p + 2$, with leading error $-(kd)^4/720$ at $p = 1$, for incident P, SV and SH waves at normal and oblique incidence, and one plane of it reproduces the

thin-layer series through $O(D^{2p+2})$. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. A smoothly graded sphere converges at second order against its exact solution. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. Where the medium varies within a cell, that sampling is itself an approximation: §9 shows that it caps any voxel at second order, unless the cell’s contrast is expanded to the degree of its field. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that hap-

17 pens with the most natural choice of coupling, the Green's tensor evaluated
 18 between voxel centres: it carries a local term whose size per unit volume does
 19 not shrink as the voxels do, and unless the single voxel's response is matched
 20 to cancel it, refinement converges to the wrong answer without revealing it.

21 *Why a homogeneous layer.* This paper therefore starts from the simplest
 22 configuration in which a voxel scheme runs in full (every voxel, every coupling,
 23 the whole multiple-scattering series, at normal and at oblique incidence) and
 24 the exact answer is nevertheless known: a layer filled with identical voxels.
 25 Because cubes tile space, such a layer does not approximate a homogeneous
 26 layer; it *is* one. Any difference between its reflection and transmission and
 27 those of the continuous layer is therefore error of the discretisation alone, with
 28 no modelling error and no shape error behind which to hide. A sphere, the
 29 usual test object for volume-integral and discrete-dipole methods [1], cannot
 30 give this: its error mixes the discretisation with the staircase approximation
 31 of a curved surface by cubes. The layer plays the part of the patch test in
 32 finite elements: reproducing a uniform medium correctly is necessary, though
 33 not sufficient, for a scheme to be trusted in a heterogeneous one. What the
 34 layer establishes about the scheme's two ingredients, the coupling between
 35 voxels and the response of a single voxel, are properties of the ingredients,
 36 which are the same in any medium: that the averaged coupling is exact, that
 37 the self term cancels, and the form of the single-site error. §9 confirms that
 38 they carry over to a stratified medium; a laterally heterogeneous one remains
 39 to be tested.

40 *What this paper adds.* Set against the coupled-dipole method described be-
 41 low, the contributions are these. (1) *An exact coupling.* The integrated

42 Green tensor and the filtered coupled dipoles approximate the field of a cell;
 43 for cubes the source-cell average is exact, because the cube's form factor van-
 44 ishes on the reciprocal lattice (the tiling identity, §5), and at normal incidence
 45 the lattice kernel is in closed form with no lattice sum. (2) *No polarisability*
 46 *to choose.* The self term the single site carries is the one the averaged kernel
 47 omits, so the discrete system is a collocation of the continuum equation (§6).
 48 The polarisability, a modelling choice in the coupled-dipole method, is here
 49 a derived object: the moments that derive it, taken as distributions, yield
 50 Eshelby's δ term without adding it by hand, the cube's departure from the
 51 sphere as a pure cubic anisotropy, and the closure's error against exact scat-
 52 tering by a sphere as a real $(ka)^2$ term in closed form. (3) *The error in closed*
 53 *form, for every grid:* $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission,
 54 so that a grid can be chosen for a required accuracy before the calculation
 55 is run; and the opposite signs, which explain why no correction of the single
 56 site can improve both (§11). (4) *Higher order.* A voxel carrying Legendre
 57 moments of its field to degree p converges at order $2p+2$ with a closed-form
 58 constant; its first-moment case converges at fourth order for P, SV and SH
 59 waves at normal and oblique incidence and in a randomly stratified model
 60 (§9), and for a reflection accuracy of 10^{-6} it allows voxels 58 times larger in
 61 each direction than the uniform one (§11). Where the medium varies within
 62 a cell, with the contrast expanded to degree r , the order is $\min(2p+2, 2r+2)$.
 63 (5) *The elastic state.* A voxel carries a force and a moment, so its state has
 64 nine components, the coupling is a 9×9 tensor, and its local term is the
 65 depolarisation of a plate. (6) *A sphere without a staircase,* whose contrast
 66 falls smoothly to zero at its surface, so that the scheme can be tested against

an exact three-dimensional solution: it converges at second order here (§10) and at fourth order in the companion paper [2].

Why cubic voxels. Cubes are the natural cell for three reasons. They *tile space*, so a uniform region discretised into cubes is still that uniform region, whereas spheres or point scatterers can only approximate it and carry a packing error of their own. Their *regular lattice* makes the interaction between cells translation invariant, so the coupling is a convolution (Toeplitz between planes, Bloch-periodic within them), and the whole system is solved by fast transforms. And a cube small compared with the wavelength has a *single-site response in closed form*, the elastic depolarisation of a cube (§6), so the only approximation a voxel introduces is the internal field it assumes.

The coupled-dipole method. Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green’s function is the multiple-scattering description of Foldy [3] and Lax [4]. In electromagnetics it became a numerical method as the coupled-dipole method, or discrete dipole approximation (DDA): each cell of a cubic grid is a polarisable point dipole driven by the incident field and by the fields of every other dipole [5, 6]. The method is a collocation of the volume integral equation for the electric field, with the cell’s self-interaction absorbed into its polarisability [7], and Yurkin and Hoekstra [8] review it in that form. Its accuracy turns on two ingredients, and this paper takes up both for elasticity. The first is the polarisability. The Clausius–Mossotti value is the static response of a sphere; the lattice dispersion relation of Draine and Goodman [9] adjusts it so that an infinite lattice propagates waves as the continuum does to a given order in kd ; Rahmani et al. [10] instead derive it from the exact static solution for

92 the whole body, so that the method is exact in the long-wavelength limit.
 93 The second is the coupling. Integrating the Green tensor over the source
 94 cell rather than sampling it at the centre improves the accuracy, most of
 95 all for strong contrasts [11], and the filtered coupled dipoles of Piller and
 96 Martin [12], revived by Yurkin et al. [13] for large refractive indices, replace
 97 the point dipole by a source of finite extent whose spectrum is band-limited
 98 to the grid. The coupling summed over a lattice extends the method to
 99 periodic structures [14]. Its error converges at second order in the dipole
 100 spacing when shape errors are absent [1], and for a cube, the one shape
 101 a cubic grid represents without a staircase, Yurkin and Kahnert [15] reach
 102 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. The layer
 103 studied here is the elastic counterpart of that cube: a body the grid represents
 104 exactly, so that the discretisation error can be isolated. What this paper adds
 105 is the closed form: for cells that tile space, the coupling that makes the lattice
 106 reproduce the continuum exactly, and the order and the constant of the error
 107 that remains.

108 *Its elastic counterparts.* The elastic problem has the same volume integral
 109 equation [16], and every cell-based solver of it is a coupled-dipole method
 110 whether or not it carries the name. Nestor [17] divided a two-and-a-half-
 111 dimensional medium into cells that interact through the background Green's
 112 tensor, each carrying a contrast renormalised by its own near field, with
 113 the multiple-scattering series summed. Kanaun and Levin [18] discretise the
 114 inclusion problem with Gaussian approximating functions whose matrix ele-
 115 ments are analytic. Yang et al. [19] solve a stress-velocity form of the integral
 116 equation on a grid by conjugate gradients with fast Fourier transforms, and

117 Touhei [20] does the same for a half space with a generalised Fourier trans-
 118 form. Jakobsen [21] and Jakobsen et al. [22] build the seismic forward and
 119 inverse problems on the transition operator of the discretised Lippmann-
 120 Schwinger equation, in the anisotropic elastic case with a nine-component
 121 state vector. The elastic problem adds a structure the scalar and electro-
 122 magnetic ones lack. A cubic voxel responds to the incident field with a force
 123 (through its density contrast) and a moment (through its stiffness contrast),
 124 so the state at a voxel has nine components (displacement and strain), and
 125 the single site's self-interaction is the elastic depolarisation of a cube, the
 126 discrete counterpart of the inclusion tensor of Eshelby [23].

127 *This paper.* It extends the formulation of Nestor [17] to cubic voxels. §2 sets
 128 out the discrete medium and §3 the continuous layer it must reproduce. §4
 129 shows why the point coupling fails, and §5 why the source-averaged coupling
 130 succeeds exactly. §6 derives the single-site T -matrix from moment integrals,
 131 tests it against the exact sphere, and shows that with it the discrete system is
 132 a collocation of the continuum equation. §7 gives that collocation's second-
 133 order error in closed form, and §8 the fourth-order voxel, for incident P
 134 and S at normal and oblique incidence, and its generalisation to Legendre
 135 moments of any degree. §9 repeats the tests on a randomly stratified layer
 136 and on a medium that varies within a cell, §10 on a sphere, and §11 discusses
 137 what the results imply for voxel schemes in general. Every result is derived
 138 symbolically in Mathematica and checked against an independent Python
 139 implementation or an exact solution.

140 2. Setting

141 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 142 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 143 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 144 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 145 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 146 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

147 displacement followed by engineering strain. A voxel responds with a force
 148 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 149 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 150 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

151 with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the
 152 point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the
 153 receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components.
 154 In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

155 where $\Delta\mathcal{C}$ is the 6×6 contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal
 156 block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

157 *Averaging, and collocation.* Three operations in this paper are averages over
 158 a cell, and they are distinct. The *source-cell average* integrates the kernel

159 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 160 of a uniform polarisation of the source cell. Every scheme in this paper uses
 161 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 162 is imposed: at the receiving cell's centre, with the internal field of each cell
 163 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 164 *average* imposes the equation in the mean over the receiving cell instead;
 165 with a uniform internal field that is the Galerkin scheme of degree zero,
 166 called the mean-only voxel below. The two are both *uniform-field* voxels, a
 167 term that refers to the internal field and not to the test, and they differ only
 168 in the test function (a delta at the centre against the cell's indicator) and
 169 in the constant of their $O((kd)^2)$ error (Table 3). The *cell-mean contrast*,
 170 finally, is an average of the material, not of the field: where the medium
 171 varies within a cell, the contrast is projected onto the cell's basis, its mean
 172 for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language
 173 of the coupled-dipole method, the point dipole is collocation with the point
 174 kernel, the integrated Green tensor adds the source-cell average, and the
 175 filtered coupled dipoles replace the point source by one of finite extent.

176 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 177 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then
 178 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 179 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 180 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

181 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 182 join the centres of the voxels in one plane (a square lattice of pitch d), and
 183 m is the separation of the two planes in units of d (Fig. 1a,b). The question
 184 is whether, and how fast, this chain converges to the continuous layer as
 185 $n \rightarrow \infty$ at fixed D .

186 *Implementations.* Every result is derived symbolically in Mathematica and
 187 compared with an independent numerical implementation or an exact solu-
 188 tion. “The package” below is the Python implementation of the voxel scheme
 189 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 190 propagator, which serve both as the object under test and, where validated,
 191 as a second implementation.

192 3. The continuous layer and its thin-layer expansion

193 At normal incidence a plane force drives a one-dimensional problem per
 194 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 195 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 196 follow from continuity of displacement and traction at its faces; they agree
 197 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 198 P and S (`ContinuumLimit_Reference.wl`).

199 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 200 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 201 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer’s wavenumber. The reflection coeffi-

202 cient expands as

$$\begin{aligned}
R = & \frac{ik_1 D}{2}(\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4}(\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24}(\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) \\
& + \frac{(k_1 D)^4}{48}(\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5),
\end{aligned}
\tag{5}$$

203 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
204 coefficient symbolically and the truncated series against the exact layer).
205 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \tag{6}$$

206 to first order a thin layer is an interface across which the traction jumps by
207 the excess inertia and the displacement by the excess compliance. At oblique
208 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
209 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
210 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
211 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
212 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
213 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
214 a single plane of voxels must reproduce order by order. A plane of voxels
215 that carry Legendre moments of their internal field to degree p reproduces
216 Eq. (5) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
217 mean-only voxel reproduces the equivalent interface and the D^2 term, the
218 first-moment voxel all four terms shown.

219 4. The specular sums of the point Green's tensor

220 *Why transform the sum.* The specular sum (4) adds, at a receiver, the field
 221 of every voxel of a plane m planes away. In that form it is a poor object to
 222 study. The Green's tensor decays slowly with distance, so the sum converges
 223 slowly, and it hides the one question that matters here: which part of it is
 224 the continuum, and which part is the discretisation. The Poisson summation
 225 formula [24] answers both at once, and is the standard tool for lattice sums
 226 of wave fields [25]. For a function f on the plane and the square lattice of
 227 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (7)$$

228 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 229 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 2).
 230 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (8)$$

231 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 232 between planes a distance z apart. Each term has a physical meaning. Trans-
 233 formed laterally, the field of a plane of sources is a superposition of plane
 234 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 235 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 236 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 237 far beyond the wavenumber of the field when the voxels are small compared
 238 with the wavelength, so it is evanescent and decays away from the plane as

239 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 240 the lattice sum exactly into the continuum and a discretisation correction,
 241 and the correction converges after a few orders.

242 *The local term that survives.* In the strain-from-moment block the evanescent
 243 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 244 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 245 term (`ContinuumLimit_SpecularSums.wl`, matched to the package’s Ewald
 246 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 247 planes in slab order, the static lattice sum of the point kernel is exactly
 248 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 249 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 250 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 251 reproduces the continuum only if that remainder is cancelled by the voxel’s
 252 own self term.

253 5. The tiling identity

254 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$
 255 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 256 Each Poisson order in Eq. (8) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (9)$$

257 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 258 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 259 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube’s indicator χ : a partition of unity (Fig. 3).
 260 Two exact statements follow.

261 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (10)$$

262 the continuum's plane-wave term times the vertical form factor. The pack-
 263 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 264 `AveragedSums.wl`).

265 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 266 is the field at the mid-plane of a uniform plate of thickness d (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (11)$$

267 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 268 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 269 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 270 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
 271 $-1/(2\mu)$ are the plate depolarisation of §4. Statics the identity holds for
 272 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 273 with every cell integral evaluated analytically (below).

274 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 275 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (12)$$

276 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 277 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 278 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}

at its default settings, falling to 2.6×10^{-5} only when both the shell radius and the quadrature order are raised; Eq. (12) is now the package's kernel at normal incidence.

Cell integrals without quadrature. Write the static (Kelvin) tensor as $G_{bc} = (c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$. Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$, which never contain the origin, and there the two-dimensional antiderivative in the face coordinates is continuous, so the face integral is a sum over the face's corners. The distributional term at $r = 0$ is carried by the divergence theorem itself. (A triple antiderivative evaluated at the cube's corners is *not* usable for repeated derivatives: its arctangent terms jump across the coordinate planes through the origin, and differentiating twice there discards exactly that term.) The self term so obtained reproduces the classical closed form of the cube's depolarisation to machine precision.

6. The single site: the cube T -matrix from moment integrals

Every error of a voxel scheme is made at the single site (§11), so the single site is derived here rather than imported. The classical route imports it. For an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a uniform load uniform, with a depolarisation tensor in closed form [23, 26, 27]; its dynamic generalisation is known for spheres and ellipsoids [28, 29]. For a cube or a general polyhedron the static field of a uniform eigenstrain is also in closed form [30, 31, 32, 33], but the strain it produces is not uniform, so there is no exact uniform-strain inclusion to import. In scattering theory

303 the same object appears as the self-consistent or effective-field closure of
 304 the volume integral equation [34, 16, 35]: a first-order (Born) scatterer [36]
 305 renormalised by the geometric series of its own near field, the step that turns
 306 a contrast into an effective contrast. Nestor [17] introduced the single site
 307 in this form, summing the series of the cell's self integral, with the singular
 308 near field evaluated over an infinitesimal cell. The moments below evaluate
 309 the same self integral over the finite cube, as a distribution. The elastic T -
 310 matrix of Waterman [37] represents a scatterer in spherical multipoles, the
 311 natural basis for a sphere; a voxel in a lattice needs one compatible with the
 312 lattice, and the moments over the cube provide it. The construction below
 313 makes the approximation of a voxel explicit rather than implicit: it derives
 314 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 315 over the cube, and identifies the uniform-strain closure as one truncation of
 316 an exact hierarchy.

317 *6.1. The closed hierarchy*

318 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda\delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 319 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 320 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 321 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed

322 system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (13)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (14)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (15)$$

323 all fields evaluated at the centre. Its coefficients are six moments of the
324 background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (16)$$

325 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
326 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
327 it, so it is fixed by convention. The moments are graded by (D, W) , the
328 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
329 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

330 6.2. The moments as distributions

331 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (17)$$

332 [38, 39]. Expanding g_c in powers of r reduces every moment in Eq. (16) to a
333 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (18)$$

334 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 335 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 336 gral depends on the shape of the region excluded around the origin. This is
 337 the source of the δ -function term of the inclusion problem, which a numerical
 338 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 339 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 340 are distributional from the start, and each is moved onto the voxel's faces by
 341 pairing with the test function $w\mathbf{1}_V$, using $\partial_q\mathbf{1}_V = -n_q\delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (19)$$

342 The surface term is evaluated on the faces, a distance h from the origin, where
 343 every derivative of r^m is smooth. The recursion lowers D and W together
 344 and ends on convergent integrals. The face integrals are done in closed form
 345 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (20)$$

346 which the face route satisfies exactly and a spherical excision returns as
 347 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
 348 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 349 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 350 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 351 Because every power of r is paired this way, the dynamic moments keep the
 352 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
 353 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 354 those terms, at a real $O((kh)^2)$ cost (§6.4).

355 6.3. Parity, and the nine-component block

356 The cube is centrosymmetric. A moment that couples a field quantity of
 357 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 358 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (14) contains the first gradient alone: ∂u is
 359 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 360 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 361 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 362 by Eq. (14), and *no treatment of the second-gradient block (15) can change*
 363 *it*. What Eq. (15) contributes reaches the far field only through the $O((kh)^2)$
 364 coupling in Eq. (13). That is the order, and the origin (the variation of the
 365 field across the voxel), of the collocation error of §7, which the first-moment
 366 voxel of §8 removes.

367 With $\Delta\rho = 0$, Eq. (14) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 368 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 369 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 370 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (21)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (22)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (23)$$

371 The density channel is the displacement block of Eq. (13) with the second
 372 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 373 of $\mathbf{G}$.

374 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
375 tional channel $1 + \Delta K / (\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
376 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
377 latational channel, Eq. (21), is the sphere's exactly: its eigenvalue involves
378 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
379 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
380 the shape. The cube's shear channels differ from each other. Their isotropic
381 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (24)$$

382 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
383 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
384 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
385 and the diagonal shear is the more depolarised. At the background of §7
386 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
387 0.4960.

388 *Where Eshelby enters.* Equations (21)–(23) follow from the moment integrals
389 and the group theory alone. No ellipsoidal result is used, and no δ term is
390 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
391 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
392 the uniformity theorem that ansatz is exact if and only if the inclusion is
393 an ellipsoid. For a sphere the construction is therefore exact in the static
394 limit, and for a cube it is an approximation. The hierarchy (13)–(15) states
395 that approximation as the omitted blocks, rather than as an unquantified
396 shape correction. A polynomial expansion about the centre cannot represent

397 the edge and corner concentrations of an isolated cube's internal field at any
398 finite degree. A voxel in a uniform layer has no such concentrations: its
399 neighbours have the same properties, so its internal field is the continuum's,
400 which is smooth. That is why a low-degree internal field suffices in a lattice
401 (§§7–8) although it is only an approximation for an isolated cube.

402 *Against the exact sphere.* These statements are tested against the exact elas-
403 tic Mie solution for a sphere, with no Eshelby formula in between. The
404 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
405 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
406 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
407 route of Eq. (19) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
408 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
409 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
410 isotropic shear average, Eq. (24), is the depolarisation read from Mie's E_2
411 to 10^{-8} . The same moment taken as a principal value, over the ball with
412 a small sphere excluded around the origin, has zero depolarisation in every
413 channel. The whole static response of the sphere is the δ term.

414 At finite frequency the closure uses the dynamic ball moments, which are
415 in closed form. It departs from Mie at second order in ka and first order in
416 the contrast, and the departure is closed-form too: the ball's moments are
417 elementary, and Mie's low-frequency series with the contrast kept exact gives

418 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (25)$$

419 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 420 the radiation damping, agree exactly in all three channels. So the closure's
 421 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 422 own. That is the order the hierarchy assigns to the omitted second gradi-
 423 ents (§6.3), and it has the same origin, the variation of the field across the
 424 scatterer, as the collocation error of §7. Eq. (25) is derived symbolically
 425 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 426 the numerical Mie solve and an independent implementation of the closure
 427 to 5×10^{-4} , which is the size of the next order in the contrast.

428 6.4. The collocation closure

429 The collocation closure takes the internal field of a voxel as uniform, the
 430 truncation of §6.3 for the density and stiffness channels together. The voxel's
 431 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 432 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 433 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (26)$$

434 With the kernel (12) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 435 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (27)$$

436 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 437 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (10), (11)),
 438 so Eq. (27) is precisely the collocation of the continuum Lippmann–Schwinger
 439 equation with a strain piecewise constant over voxels. It does not contain
 440 $\mathbf{S}_{\text{self}}$ at all.

441 The package’s cube T -matrix is this closure statically: both blocks agree
 442 with Eq. (26) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 443 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about $0.25(kh)^2$
 444 through its far-field form-factor corrections. Its self integrals are exact in the
 445 static and radiation parts but omit the real even-power terms of the Green’s
 446 tensor’s expansion ($k^2 r$, r^3 , $\dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 447 self term, whose series agrees with the closed-form elastodynamic tensor to
 448 10^{-14} . By Eq. (27) that error cannot enter the closure scheme.

449 7. Convergence, and the closed-form error

450 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 451 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 452 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 453 the chain converges to the continuous layer’s at order 2.000 for $n = 4 \rightarrow$
 454 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (27) and with the
 455 package T -matrix alike (`ContinuumLimit_Chain.wl`). The error depends on

456 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
 457 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
 458 the exact solution and n the refinement; the orders in the legend of Fig. 4
 459 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
 460 with their standard errors. The two agree where a sequence is asymptotic
 461 and differ where it is not, which the apparent orders show.

462 *The leading error in closed form.* In Eq. (27) the kernel is integrated exactly;
 463 the only approximation is the internal field's variation across a voxel. At first
 464 order in the contrast the discrete scattered field is therefore a midpoint rule
 465 applied to the continuum's Born integral, with the kernel's cell integral exact.
 466 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 467 it is flat. With k the background wavenumber of the wave concerned (k_P for
 468 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 469 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (28)$$

470 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 471 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 472 and those of Eq. (31) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 473 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 474 the transmission constant grows to -0.092 .

475 *Why a uniform-field single site stops at second order.* The two constants have
 476 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 477 cancel both. The package *T*-matrix illustrates it: its form-factor corrections

478 reduce the reflection error by a factor 4.45 and *increase* the transmission
 479 error by 1.7, at every n . The missing term in Eq. (28) is the first moment
 480 of the internal field over each voxel, which radiates through the derivative of
 481 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 482 channel, and a kernel whose cell integrals include the kernel's derivative.

483 8. Fourth order and beyond: Legendre moments of the voxel field

484 Let each voxel carry the mean *and* the first moment of its internal field.
 485 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
 486 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
 487 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the
 488 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
 489 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (29)$$

490 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (30)$$

491 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
 492 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
 493 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
 494 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
 495 verges at fourth order in both reflection and transmission (`ContinuumLimit_`

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (30). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

496 **FourthOrder.w1**; Table 1). A single plane of first-moment voxels represents
497 the 2 m layer 6500 times more accurately than a single plane of uniform ones.
498 The collocation column is the scheme of §6 and reproduces the chain of the
499 full 9×9 system to all printed digits.

500 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
501 ing in the construction: the moments and the same-cell double integrals of
502 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-
503 degree solver reproduces the chain above exactly. The second-moment voxel
504 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
505 eighth, in reflection and transmission (Table 2; **ContinuumLimit_SecondMoment.**
506 **w1**); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
507 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

508 *The leading error in closed form.* Without scattering, the Galerkin solution
509 is the L^2 projection Π of the incident field onto each voxel's basis, and the
510 kernel is integrated over each voxel exactly; at first order in the contrast the

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} , $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (31)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (30) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6} . The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 3: The leading error of the discrete layer in closed form, Eq. (31): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

525 $h = d/2$,

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (32)$$

526 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
527 mission for odd p , opposite signs for even p . The first-moment voxel's leading
528 error therefore does not depend on the direction in which the field is observed.
529 Eq. (32) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
530 ishes) in Mathematica and, independently, in sympy.

531 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
532 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
533 $d = D$) the series in D of the reflection coefficient, computed exactly at the
534 full contrast, coincides with that of the exact layer, Eq. (5), term by term
535 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
536 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

537 `w1`). Each further Legendre moment adds two terms of the thin-layer series.

538 *In three dimensions.* The three-dimensional first-moment voxel carries the
 539 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 540 in all, tested with the same functions and coupled by cell-to-cell double in-
 541 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 542 cidence it reduces *exactly* to Eq. (30) (`ContinuumLimit_Reduction3D.w1`).
 543 The lateral first moments are never excited: the lattice and the incident
 544 field are symmetric under reflection about every voxel's centre planes, and
 545 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 546 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 547 ishes on the reciprocal lattice and the plane sum of the double integrals
 548 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 549 three-dimensional structure begins. The lateral first moment's form fac-
 550 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 551 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 552 evanescent orders. And at oblique incidence the mean's own zeros shift,
 553 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 554 only to $O(k_{\parallel} h)$.

555 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
 556 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g},$$

(33)

557 with L_a the lateral Legendre factors. The whole-space tensor decays as
 558 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each

entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma + i\beta \operatorname{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (33). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (30) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (30) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full 9×9 construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique inci-

dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

575

576 of Eqs. (28) and (31), are those of the incident P wave.

577 The spectral construction is implemented twice. The second implementa-
578 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
579 nition of the scheme: it takes the depth dependence of the transformed kernel
580 from an independent residue construction, obtains the local weight as a nu-
581 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
582 ble integral by quadrature and separates P from S by sampling the upgoing
583 field at two depths. The raw reflected displacements of the two agree to $9 \times$
584 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
585 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
586 scheme of any degree is also implemented twice. The second implementation
587 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
588 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
589 from its own interface system, in double precision. Its errors agree with the
590 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in

591 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 592 6.00; for one plane as the layer thins its differences from the exact layer fall
 593 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 594 Figure 4(a,b) shows the convergence, with the orders fitted to it.

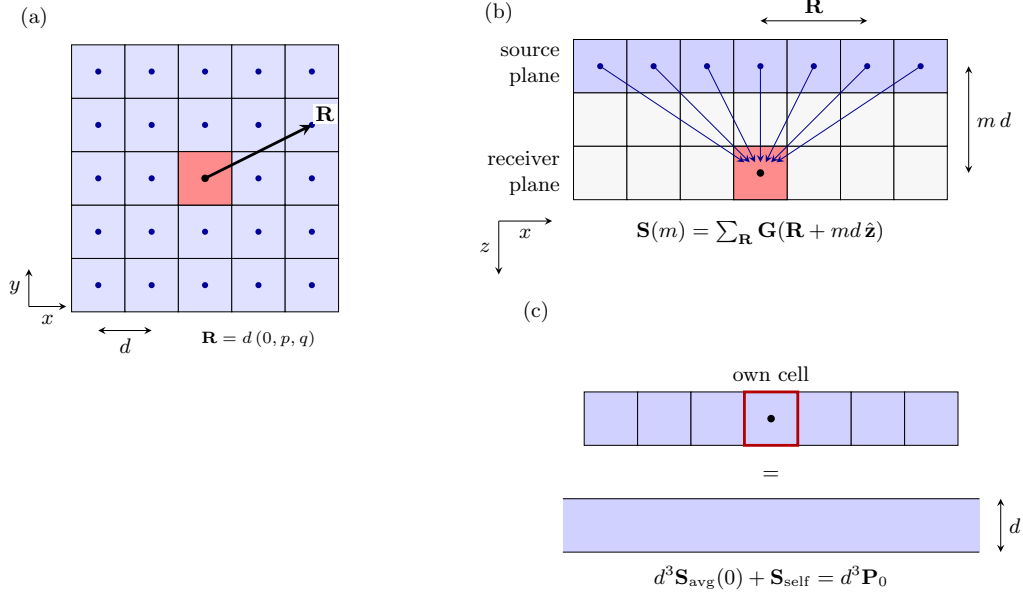


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (4). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (11).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

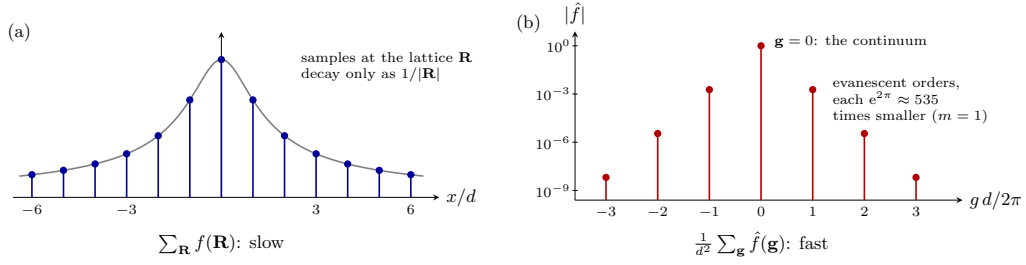


Figure 2: The Poisson summation formula (7) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

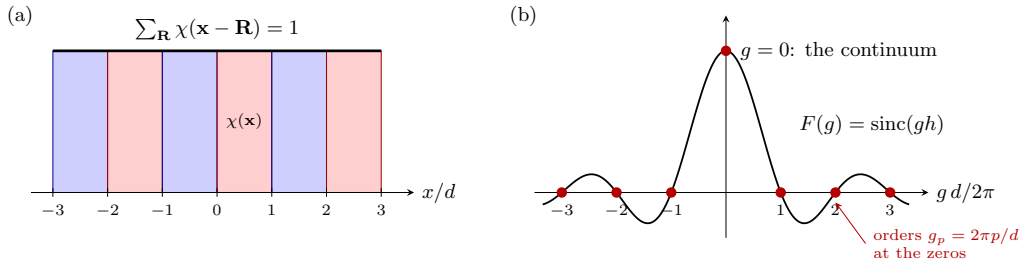


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (8) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

595 9. A heterogeneous layer

596 A heterogeneous medium is not a layer of identical cells. The simplest
597 heterogeneous case with an exact answer is a stratified model built from layers
598 of constant properties, with real interfaces between them: the form of many
599 layered crustal velocity models. The last test takes such a model. A layer
600 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
601 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
602 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
603 stratification with physical contrasts. The voxels refine the model's layering,
604 with m planes of voxels in each model layer, so no voxel straddles a model
605 interface: the natural discretisation when the model's interfaces are known.
606 At $\omega = 1500$ rad s⁻¹ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
607 reverberates between the model interfaces. The exact answer is the product
608 of the model layers' propagators. Its same-type reflection coefficients agree
609 with those of the reflection–transmission recursion of Kennett [40], computed
610 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
611 now changing from plane to plane; nothing else changes. The errors are in
612 Table 6 and Fig. 4(c).

613 *The interfaces cost nothing.* Every order is the one measured for the uniform
614 layer. At a model interface the strain of the true field jumps. A voxel basis
615 that is discontinuous from voxel to voxel represents that jump exactly when
616 the interface lies on a voxel boundary, so the interface adds no error of its
617 own. The only requirement is that voxel boundaries coincide with the model's
618 interfaces. A voxel that straddled an interface would carry a contrast that is
619 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

620 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
621 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
622 layer. The first-moment voxel needs 8, one per model layer: the model's own
623 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
624 against 36), that is 3123 against 288 per vertical column of voxels, a factor
625 of eleven. In three dimensions, with the lateral grid refined as the vertical
626 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
627 unknowns by about 2×10^4 . The same model has been solved independently
628 in Python, with a contrast for each plane: the raw reflected displacements
629 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
630 at 20° , with and without the first moments. Lateral heterogeneity needs the
631 full three-dimensional solver; it is the first extension discussed below.

632 *A medium that varies within a cell.* The stratified model is exact on its own
 633 grid: every cell is homogeneous. A smoothly varying medium is not, and
 634 sampling it as cells of constant properties is an approximation of its own.
 635 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
 636 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
 637 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
 638 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (30)
 639 becomes

$$\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\int_{\mathbf{i}} \int_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (34)$$

640 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 641 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 642 the order of convergence is

$$\min(2p+2, 2r+2), \quad (35)$$

643 or $2p+2$ when degree r represents the profile exactly (Table 7; `ContinuumLimit_`
 644 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 645 caps every voxel at second order, however rich its field basis, and for the
 646 smooth profile the voxel of degree 2 with a linear contrast has the error
 647 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 648 this is what one expects: the contrast's projection error is orthogonal to
 649 the polynomials of degree r , so against the smooth product of the inci-
 650 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 651 the contrast must be represented to the degree of the field. The closed forms
 652 of the construction are tested one by one by an independent implementa-

Table 7: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (35), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
bles the scheme without linearisation, evaluating the projected contrast at
the quadrature nodes, against its own 30-digit solution of the graded layer;
its errors agree with those of Table 7 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it
measures the orders of Eq. (35) again.

10. A sphere: what the layer predicts in three dimensions

The layer and the stratified layer are laterally uniform. A sphere is not,
and elastic scattering by a sphere, homogeneous [41, 42] or radially varying
[43], has an exact solution: it is the usual benchmark for volume-integral and

665 discrete-dipole methods. This section solves two spheres with the full three-
 666 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
 667 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
 668 iterative solver, and a P wave is incident. For the graded sphere the voxel is
 669 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
 670 nine-component state, tested with the same constant, and the L^2 projection
 671 of its contrast, which is its cell mean; every cell that overlaps the sphere is
 672 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
 673 tensor, one block per orbit of the cube group [2]. For the sharp sphere
 674 the voxel is the package's cube T -matrix, which differs from the collocation
 675 closure only through its far-field corrections (§6.4) and has the same order,
 676 each cell keeping the contrast of its centre. The far field is computed from
 677 the solution for the incident plane wave itself, each voxel's displacement and
 678 strain carrying the plane-wave phase at its centre; adding to that phase a
 679 linear variation about the scatterer's centre, as a Taylor expansion of the
 680 incident field would, counts the wave's variation twice and leaves an error of
 681 order $(ka)^2$ that no refinement removes. The error is the largest difference
 682 between the scattered far field and the exact one over nine scattering angles
 683 from 11° to 169° , relative to the largest exact value, at a distance of 5×10^8
 684 radii for the graded sphere: at 5×10^4 radii the near-field terms that the
 685 asymptotic far field omits are 1.4×10^{-4} of the peak, 4% of the discretisation
 686 error at 12 voxels across and most of it at 24. The sharp sphere, whose errors
 687 are a hundred times larger, is observed at 5×10^4 radii.

688 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
 689 its shape with every grid. A sphere whose contrast falls smoothly to zero at

690 its surface does not: the voxels that cross the surface carry a vanishing con-
 691 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
 692 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
 693 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes
 694 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 695 the radial equations of Takeuchi and Saito [43], integrated across the shell
 696 from the core's regular solutions and matched to outgoing waves at $r = a$.
 697 The equations are derived symbolically from the equations of motion, and
 698 the solution is checked against a sphere of homogeneous concentric shells
 699 in closed form, by unitarity and reciprocity, and by an independent imple-
 700 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 701 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 702 carries the mean of the contrast over its cell. Table 8 shows the error falling
 703 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/pilot_`
 704 `graded_voxel_sphere.py`): the apparent order between successive grids is
 705 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders
 706 of Fig. 4(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
 707 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 708 graded shell and the order is not yet asymptotic. This is the order the layer
 709 predicts for the mean-only voxel, and the first test of that prediction on a
 710 body with genuine three-dimensional structure.

711 What makes the sequence this clean is that the voxel carries a single error.
 712 The collocation voxel of the sharp sphere applied to this body, each cell at
 713 the profile's value at its centre and only the cells whose centres lie inside the
 714 sphere kept, carries two: its own internal-field error and the error of sampling

715 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
 716 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
 717 2.0, 3.5 and 3.6 at $k_{Sa} = 0.5$, and at $k_{Sa} = 1$ the apparent orders are 1.7, 2.5
 718 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,
 719 but its apparent order is not that of any single term and cannot be read from
 720 them. The same collocation voxel with the cell mean of the contrast, and
 721 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
 722 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
 723 second term, and with it removed the collocation and mean-only voxels differ
 724 only by their constants, as in Table 3. The cell mean, with every overlapping
 725 cell kept, represents the same body on every grid, and the order the layer
 726 predicts is then the one observed.

727 The first-moment voxel of §8, extended to three dimensions with a con-
 728 trast linear in each cell, converges at fourth order on this same sphere [2]: its
 729 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
 730 at weak contrast, and at the full contrast its error with 16 voxels across is
 731 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
 732 the same grid.

733 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 734 and its voxelisation changes with every grid. Table 9 shows its error for grids
 735 of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_{Sa} = 0.5$. The
 736 error tracks the staircase's volume, which is 5%, 17% and 4% too large on
 737 these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
 738 it falls below 1% at $k_{Sa} = 0.5$ and to at most 3% at $k_{Sa} = 1.0$, but it still
 739 does not converge: what remains is the staircase's shape. A refinement study

Table 8: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{ga}} = 0.5$		$k_{\text{ga}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

740 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
741 measures the scheme.

Table 9: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

742 11. Discussion

743 Figure 4 collects the convergence of every test, with the order fitted to
744 each by least squares (`scripts/plot_convergence_orders.py`).

745 *Where the error of a voxel scheme lives.* The results separate the two ap-
746 proximations a voxel scheme makes. The *coupling* between voxels makes
747 none: averaged over the source cell it is the continuum’s cell integral ex-
748 actly, between planes by the form factor’s zeros and within a plane by the
749 tiling identity, and the self term it omits is exactly the one the single site
750 carries. Every error therefore comes from the *single site*, from the internal
751 field it assumes. For a uniform field that error is the midpoint rule’s, known
752 in closed form; for a linear field it is the next term of the same expansion.
753 The converse is the practical warning. A kernel that is not the cell integral
754 (the point Green’s tensor, or a cell average that does not match the single
755 site’s closure) leaves a local term that does not vanish as the voxels shrink
756 (§4): the evanescent lattice orders carry a static contribution of fixed size per

757 voxel volume. A scheme built that way converges to the wrong limit, and
 758 refinement cannot reveal it, because the error is scale-invariant.

759 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 760 single site from a modelling choice into a derived object with a stated error.
 761 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 762 only by the cubic split of Eq. (24). Its dynamic part departs from exact
 763 scattering by a sphere only through the real $(ka)^2$ term of Eq. (25); the
 764 radiation damping is exact. That term and the collocation error of Eq. (28)
 765 have the same origin, the variation of the field across the scatterer, and the
 766 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 767 field a voxel must represent is the smooth continuum field, not the field of
 768 an isolated cube, with its concentrations at the edges and corners.

769 *Relation to the coupled-dipole method.* The same partition appears in the
 770 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
 771 relation, the lattice dispersion relation [9], the polarisability taken from the
 772 exact static solution [10] and the integrated Green tensor [11] all adjust
 773 the single site so that the lattice reproduces the continuum. For space-
 774 filling cubes the adjustment is neither needed nor a free parameter: the tiling
 775 identity fixes the local field exactly, the self term of the collocation closure is
 776 the one that makes it cancel, and what is left is a closed-form error rather than
 777 an empirical one. That the method is a collocation of the volume integral
 778 equation was shown for the electromagnetic case by Lakhtakia [7]; Eq. (27)
 779 is the same statement for the elastic lattice with the self term evaluated over
 780 the finite cube, and it is what makes the error a midpoint-rule error. The

781 filtered coupled dipoles of Piller and Martin [12] and Yurkin et al. [13] band-
 782 limit the source to the grid; the source-cell average does the complementary
 783 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 784 spectral construction of Eq. (33) is the periodic coupled-dipole method of
 785 Chaumet et al. [14] with the Galerkin double average in place of the point
 786 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 787 [1] corresponds to the uniform-field voxel's, and the opposite-signed constants
 788 of Eq. (28) explain why a polarisability correction can improve one observable
 789 only at the expense of another, as the package's own form-factor corrections
 790 do in reflection and transmission. Going beyond second order requires a richer
 791 internal field, which is the point of higher-order bases for volume integral
 792 equations; the first-moment voxel is the lowest such basis, and in its spectral
 793 form the Galerkin double average makes the lattice sum converge without
 794 Ewald splitting.

795 *The tension between the two error sources.* The coupled-dipole method has
 796 two sources of error, the single site and the coupling between sites, and each
 797 has been corrected on its own: the polarisability by the Clausius–Mossotti
 798 relation, the lattice dispersion relation or the exact static limit, and the
 799 coupling by integrating or filtering the Green tensor. The two are not in-
 800 dependent. The point kernel's evanescent orders carry a static local term
 801 whose size per voxel volume does not shrink with the voxels (§4), and the
 802 single site must cancel exactly that term. A polarisability tuned for one
 803 kernel is therefore wrong for another: the lattice dispersion relation corrects
 804 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 805 tered one changes the term it must cancel. Tuning either alone moves the

806 error between the two, and refinement cannot reveal a mismatch, because
 807 the surviving term is scale-invariant. For space-filling cubes the tension is
 808 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 809 term that the closure carries (Eq. (27)), so the coupling error vanishes iden-
 810 tically and every remaining error is the single site's, the variation of the field
 811 across the cell. That error has a second tension of its own. It is a midpoint-
 812 rule error whose constants have opposite signs in reflection and transmission
 813 (Eq. (28)), so any scalar adjustment of the single site that improves one wors-
 814 ens the other; the package's far-field corrections reduce the reflection error
 815 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 816 however chosen, can remove it. What removes it is a richer internal field:
 817 the first moment lowers the error by two orders in kd in both observables at
 818 once.

819 *The cost of fourth order.* A first-moment voxel carries four times as many
 820 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 821 forms show what that buys. To reach an error ε in reflection, the uniform
 822 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 823 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 824 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 825 number of unknowns by about 5×10^4 (these are the Born-order constants;
 826 at the contrast studied the first-moment voxel's measured errors lie within
 827 5% of them, while the uniform voxel's transmission constant grows with the
 828 contrast, so the comparison is if anything conservative). The stratified model
 829 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 830 moment voxels for the same accuracy. Higher moments continue the trend

831 in one dimension: by Eq. (32) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 832 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 833 a quarter of a wavelength. A three-dimensional voxel of degree p has not
 834 been built; with the Legendre polynomials of total degree at most p it would
 835 carry 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for
 836 the first-moment voxel. The gain is largest precisely where it matters: when
 837 the model varies on scales that are coarse compared with the accuracy the
 838 wavefield must have.

839 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 840 mogeneous layer and a randomly stratified one, for which the discrete system
 841 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 842 moment voxel in a full three-dimensional solver, where the gradient channels
 843 couple through the evanescent lattice orders (§8) and the tiling identity holds
 844 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 845 voxel converging at second order on a laterally heterogeneous body against
 846 an exact solution, and the companion paper [2] shows the first-moment voxel
 847 converging at fourth order there. A contrast that varies within a cell, which
 848 §9 shows is needed beyond second order in a smoothly varying medium, has
 849 a further consequence in three dimensions: its linear part is odd about the
 850 cell's centre, so it couples the single site's uniform-field and first-moment
 851 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
 852 voxel therefore needs at least the first-moment block; that voxel is built,
 853 and tested on the graded sphere, in the companion paper. The numerical
 854 evidence is at moderate, physically typical contrasts (up to 15% in density
 855 and moduli) and at $k_{\parallel}h \leq 0.3$, the range in which the single-site T -matrix is

856 validated; the closed forms are first order in the contrast, and the departures
 857 from them at the full contrast are of second order. What is established is the
 858 *order* of convergence and its leading constant, not the behaviour at strong
 859 contrast, where the single site's own closure may limit the accuracy before
 860 the lattice coupling does. The single site's dynamic error is in closed form
 861 for a sphere only; for a cube, for which no exact solution exists, it is known
 862 through the layer (§§7–8). Two extensions follow directly. The first is the
 863 three-dimensional solver itself. The second is a stratified *background*, a lay-
 864 ered reference medium, as distinct from the stratified contrast of §9. There
 865 the lattice coupling then carries the reverberation of the layering, which is
 866 not translation invariant in depth, and whether a tiling identity holds for it
 867 is the natural next question.

868 12. Conclusions

869 A lattice of cubic voxels reproduces a continuous elastic medium in a pre-
 870 cise sense. Because cubes tile the plane, the source-cell-averaged coupling be-
 871 tween voxels is the continuum's cell integral exactly, and at normal incidence
 872 the lattice kernel is available in closed form with no lattice sum. The single
 873 site is derived rather than assumed. The moments of the Green's tensor over
 874 the cube, taken as distributions, give a closed hierarchy whose first-gradient
 875 truncation is the uniform-strain closure. That closure is exact for a sphere in
 876 the static limit, and it departs from exact scattering by a sphere only by a
 877 real $(ka)^2$ term, with the radiation damping exact. For a cube it differs from
 878 the sphere only by a cubic anisotropy of the shear response. The collocation
 879 closure carries the same self term as the lattice kernel, which cancels, so the

880 discrete system is a collocation of the continuum equation. It converges at
 881 second order, with a leading error that is a midpoint-rule error of the internal
 882 field, known in closed form for every discretisation, and opposite in sign in
 883 reflection and transmission. A voxel that also carries the first moment of
 884 its internal field converges at fourth order, at normal incidence (where the
 885 three-dimensional scheme reduces exactly to one dimension) and at oblique
 886 incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the
 887 direction of observation. With Legendre moments to degree p it converges at
 888 order $2p + 2$ with the closed-form constant of Eq. (32), and a single plane of
 889 such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$.
 890 Where the medium varies within a cell, a contrast sampled as constant caps
 891 every voxel at second order; with the contrast projected to degree r the order
 892 is $\min(2p + 2, 2r + 2)$, and the coupling needs nothing beyond the same kernel
 893 moments to degree $p + r$. The orders survive the interfaces of a randomly
 894 stratified model, provided voxel boundaries coincide with its interfaces, and
 895 there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel
 896 per model layer, where the uniform-field voxel needs about forty. A sphere
 897 whose contrast falls smoothly to zero at its surface converges at second order
 898 against its exact solution, as the layer predicts, to 0.018% at $k_S a = 0.5$ with
 899 twenty-four voxels across; a sharp sphere's error is set by its staircase, which
 900 changes with the grid. The results, and the evidence for each, are collected
 901 in Table 10.

Table 10: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (5)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (11)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (12)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (19)	sum rule (20) exact; M, Q equal an independent archive, 81/81
Channels (21)–(23)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (24)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (25)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (27)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (28)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (30)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (33)	order 4, P and S, 20° and 40°

902 **Acknowledgments**

903 This work began thirty years ago in the author’s doctoral research at
904 the Research School of Earth Sciences of the Australian National University,
905 Canberra [17].

906 **Declaration of competing interest**

907 The author declares no known competing financial interests or personal
908 relationships that could have appeared to influence the work reported in this
909 paper.

910 **Declaration of generative AI and AI-assisted technologies in the** 911 **manuscript preparation process**

912 During the preparation of this work, the author used Claude (Anthropic)
913 in order to check the existing Mathematica notebooks, Python and Fortran
914 code, and to build tests to ensure their correctness. After using this tool, the
915 author reviewed and edited the content as needed and takes full responsibility
916 for the content of the published article.

917 **Data availability**

918 The Mathematica notebooks and Python scripts that derive and check ev-
919 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
920 5281/zenodo.23003241 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
921 MultipleScatteringCalculations.

922 References

- 923 [1] M. A. Yurkin, V. P. Maltsev, A. G. Hoekstra, Convergence of the dis-
924 crete dipole approximation. I. Theoretical analysis, *J. Opt. Soc. Am. A*
925 23 (2006) 2578–2591.
- 926 [2] T. M. Nestor, Graded voxels in three dimensions: A galerkin first-
927 moment cube with a contrast linear in the cell, fourth order in its dis-
928 cretisation, 2026. Preprint, archived with this paper at [https://doi.](https://doi.org/10.5281/zenodo.23003241)
929 [org/10.5281/zenodo.23003241](https://doi.org/10.5281/zenodo.23003241).
- 930 [3] L. L. Foldy, The multiple scattering of waves. I. General theory of
931 isotropic scattering by randomly distributed scatterers, *Phys. Rev.* 67
932 (1945) 107–119.
- 933 [4] M. Lax, Multiple scattering of waves, *Rev. Mod. Phys.* 23 (1951) 287–
934 310.
- 935 [5] E. M. Purcell, C. R. Pennypacker, Scattering and absorption of light by
936 nonspherical dielectric grains, *Astrophys. J.* 186 (1973) 705–714.
- 937 [6] B. T. Draine, P. J. Flatau, Discrete-dipole approximation for scattering
938 calculations, *J. Opt. Soc. Am. A* 11 (1994) 1491–1499.
- 939 [7] A. Lakhtakia, Strong and weak forms of the method of moments and the
940 coupled dipole method for scattering of time-harmonic electromagnetic
941 fields, *Int. J. Mod. Phys. C* 3 (1992) 583–603.
- 942 [8] M. A. Yurkin, A. G. Hoekstra, The discrete dipole approximation: an

943 overview and recent developments, *J. Quant. Spectrosc. Radiat. Transfer*
944 106 (2007) 558–589.

945 [9] B. T. Draine, J. Goodman, Beyond Clausius–Mossotti: wave propaga-
946 tion on a polarizable point lattice and the discrete dipole approximation,
947 *Astrophys. J.* 405 (1993) 685–697.

948 [10] A. Rahmani, P. C. Chaumet, G. W. Bryant, Coupled dipole method
949 with an exact long-wavelength limit and improved accuracy at finite
950 frequencies, *Opt. Lett.* 27 (2002) 2118–2120.

951 [11] P. C. Chaumet, A. Sentenac, A. Rahmani, Coupled dipole method for
952 scatterers with large permittivity, *Phys. Rev. E* 70 (2004) 036606.

953 [12] N. B. Piller, O. J. F. Martin, Increasing the performance of the coupled-
954 dipole approximation: a spectral approach, *IEEE Trans. Antennas*
955 *Propag.* 46 (1998) 1126–1137.

956 [13] M. A. Yurkin, M. Min, A. G. Hoekstra, Application of the discrete
957 dipole approximation to very large refractive indices: filtered coupled
958 dipoles revived, *Phys. Rev. E* 82 (2010) 036703.

959 [14] P. C. Chaumet, A. Rahmani, G. W. Bryant, Generalization of the
960 coupled dipole method to periodic structures, *Phys. Rev. B* 67 (2003)
961 165404.

962 [15] M. A. Yurkin, M. Kahnert, Light scattering by a cube: accuracy limits of
963 the discrete dipole approximation and the T-matrix method, *J. Quant.*
964 *Spectrosc. Radiat. Transfer* 123 (2013) 176–183.

- 965 [16] J. E. Gubernatis, E. Domany, J. A. Krumhansl, Formal aspects of the
966 theory of the scattering of ultrasound by flaws in elastic materials, J.
967 Appl. Phys. 48 (1977) 2804–2811. doi:10.1063/1.324141.
- 968 [17] T. M. Nestor, Seismic Wave Propagation in Complex Media: Sum-
969 ming the Multiple Scattering Series, Ph.D. thesis, Research School of
970 Earth Sciences, Australian National University, Canberra, ACT, Aus-
971 tralia, 1996. Open access: <https://hdl.handle.net/1885/144359>.
- 972 [18] S. K. Kanaun, V. M. Levin, Scattering of elastic waves on a hetero-
973 geneous inclusion of arbitrary shape: an efficient numerical method for
974 3D-problems, Wave Motion 50 (2013) 687–707.
- 975 [19] J. Yang, A. Abubakar, P. M. van den Berg, T. M. Habashy, F. Reitich,
976 A CG-FFT approach to the solution of a stress-velocity formulation of
977 three-dimensional elastic scattering problems, J. Comput. Phys. 227
978 (2008) 10018–10039.
- 979 [20] T. Touhei, A fast volume integral equation method for elastic wave
980 propagation in a half space, Int. J. Solids Struct. 48 (2011) 3194–3208.
- 981 [21] M. Jakobsen, T-matrix approach to seismic forward modelling in the
982 acoustic approximation, Stud. Geophys. Geod. 56 (2012) 1–20.
- 983 [22] M. Jakobsen, I. Pšenčík, E. Iversen, B. Ursin, Transition operator ap-
984 proach to seismic full-waveform inversion in arbitrary anisotropic elastic
985 media, Commun. Comput. Phys. 28 (2020) 297–327.
- 986 [23] J. D. Eshelby, The determination of the elastic field of an ellipsoidal

- inclusion, and related problems, *Proc. R. Soc. Lond. A* 241 (1957) 376–396.
- [24] E. M. Stein, G. Weiss, *Introduction to Fourier Analysis on Euclidean Spaces*, Princeton University Press, Princeton, NJ, 1971.
- [25] C. M. Linton, Lattice sums for the Helmholtz equation, *SIAM Rev.* 52 (2010) 630–674. doi:10.1137/09075130X.
- [26] J. D. Eshelby, The elastic field outside an ellipsoidal inclusion, *Proc. R. Soc. Lond. A* 252 (1959) 561–569. doi:10.1098/rspa.1959.0173.
- [27] T. Mura, *Micromechanics of Defects in Solids*, 2nd ed., Martinus Nijhoff, Dordrecht, 1987. doi:10.1007/978-94-009-3489-4.
- [28] Y. Mikata, S. Nemat-Nasser, Elastic field due to a dynamically transforming spherical inclusion, *J. Appl. Mech.* 57 (1990) 845–849. doi:10.1115/1.2897650.
- [29] T. M. Michelitsch, H. Gao, V. M. Levin, Dynamic Eshelby tensor and potentials for ellipsoidal inclusions, *Proc. R. Soc. Lond. A* 459 (2003) 863–890. doi:10.1098/rspa.2002.1054.
- [30] Y. P. Chiu, On the stress field due to initial strains in a cuboid surrounded by an infinite elastic space, *J. Appl. Mech.* 44 (1977) 587–590. doi:10.1115/1.3424140.
- [31] J. K. Lee, W. C. Johnson, Calculation of the elastic strain field of a cuboidal precipitate in an anisotropic matrix, *Phys. Status Solidi A* 46 (1978) 267–272. doi:10.1002/pssa.2210460134.

- 1009 [32] G. J. Rodin, Eshelby's inclusion problem for polygons and polyhedra, J.
1010 Mech. Phys. Solids 44 (1996) 1977–1995. doi:10.1016/S0022-5096(96)
1011 00066-X.
- 1012 [33] H. Nozaki, M. Taya, Elastic fields in a polyhedral inclusion with uniform
1013 eigenstrains and related problems, J. Appl. Mech. 68 (2001) 441–452.
1014 doi:10.1115/1.1362670.
- 1015 [34] A. K. Mal, L. Knopoff, Elastic wave velocities in two-component sys-
1016 tems, J. Inst. Math. Appl. 3 (1967) 376–387. doi:10.1093/imamat/3.
1017 4.376.
- 1018 [35] S. K. Kanaun, V. M. Levin, Self-Consistent Methods for Compos-
1019 ites, Vol. 2: Wave Propagation in Heterogeneous Materials, Solid Me-
1020 chanics and Its Applications, Springer, Dordrecht, 2008. doi:10.1007/
1021 978-1-4020-6968-0.
- 1022 [36] R.-S. Wu, K. Aki, Scattering characteristics of elastic waves by an elastic
1023 heterogeneity, Geophysics 50 (1985) 582–595. doi:10.1190/1.1441934.
- 1024 [37] P. C. Waterman, Matrix theory of elastic wave scattering, J. Acoust.
1025 Soc. Am. 60 (1976) 567–580. doi:10.1121/1.381130.
- 1026 [38] V. D. Kupradze, Dynamical Problems in Elasticity, volume 3 of *Progress*
1027 *in Solid Mechanics*, North-Holland, Amsterdam, 1963.
- 1028 [39] K. Aki, P. G. Richards, Quantitative Seismology, 2nd ed., University
1029 Science Books, Sausalito, CA, 2002.

- 1030 [40] B. L. N. Kennett, *Seismic Wave Propagation in Stratified Media*, Cam-
1031 bridge University Press, Cambridge, 1983.
- 1032 [41] C. F. Ying, R. Truell, Scattering of a plane longitudinal wave by a
1033 spherical obstacle in an isotropically elastic solid, *J. Appl. Phys.* 27
1034 (1956) 1086–1097. doi:10.1063/1.1722545.
- 1035 [42] Y.-H. Pao, C.-C. Mow, *Diffraction of Elastic Waves and Dynamic Stress*
1036 *Concentrations*, Crane Russak, New York, 1973.
- 1037 [43] H. Takeuchi, M. Saito, Seismic surface waves, in: B. A. Bolt (Ed.),
1038 *Methods in Computational Physics*, volume 11, Academic Press, New
1039 York, 1972, pp. 217–295.

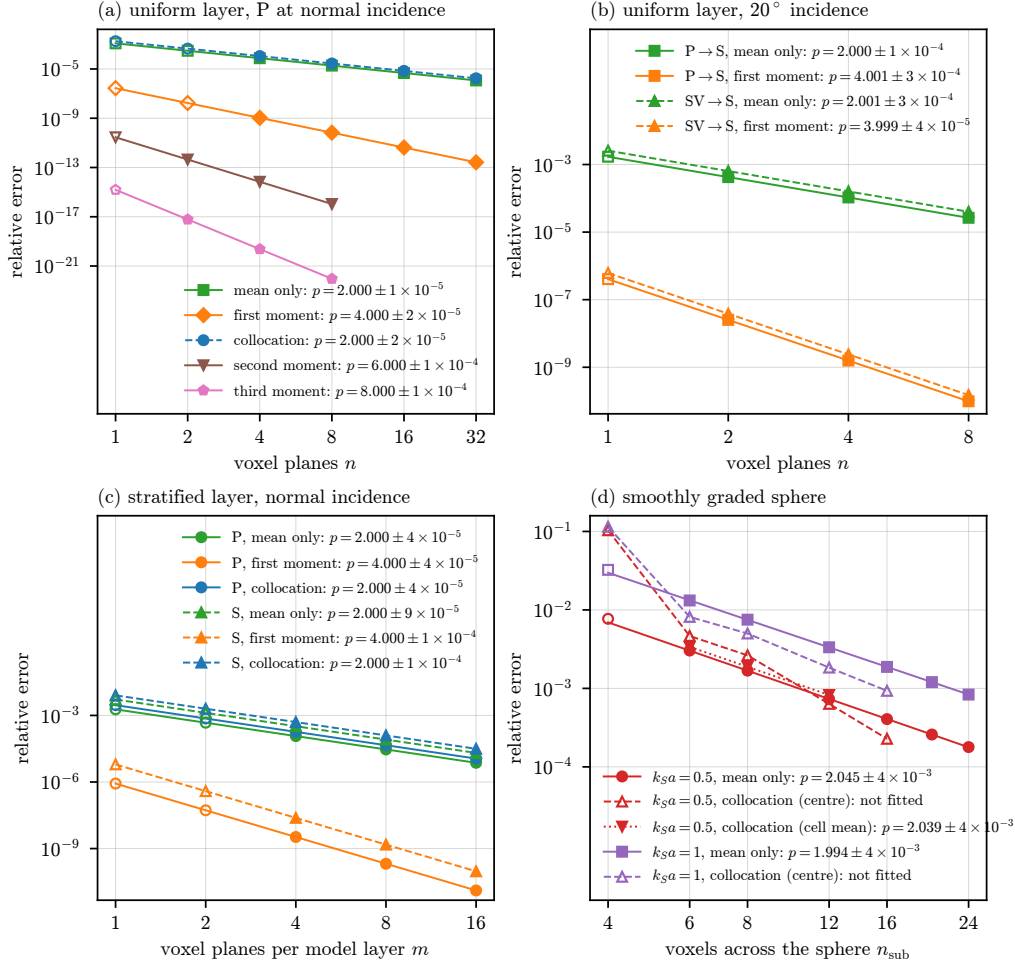


Figure 4: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree $p = 1, 2$ and 3 (Eqs. (30) and (33)). (a) The uniform layer, P reflection at normal incidence ($D = 2\text{ m}$, $\omega = 300\text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 3; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and not fitted, since its two $O(h^2)$ errors of opposite sign leave no single order to fit; and, at $k_{\text{sa}} = 0.5$, the same collocation voxel with the